

`--init__(n_clusters, max_iter)`
step 1: store `k` and the round limit

`fit_predict(X)`
step 2: `k` random rows become the centroids

`assign_clusters(X)`
step 3: each row gets the index of its nearest centroid

`move_centroids(X, cluster_group)`
step 4: each centroid = mean of its rows

step 5: centroids
unchanged?

no, and fewer
than `max_iter`
rounds so far

yes
`return cluster_group`

